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Paleomagnetic and rock magnetic study of the Mesoproterozoic sill, Valaam island, Russian Karelia

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Abstract

A paleomagnetic study of the Mesoproterozoic Valaam monzodioritic sill (U-Pb age ~1458 Ma) was undertaken in an attempt to provide a new well dated paleomagnetic pole for Baltica and to test the proposed joint drift history of Baltica and Laurentia from 1.83 to ca. 1.26 Ga. Two components of natural remanent magnetization (NRM) were isolated in the sill with alternating field and thermal treatments. The first component is a low coercivity component of viscous origin with a direction similar to that of the Earth's present field at the sampling site. The second one (component **A**) with a high unblocking temperature/coercivity has a remanent magnetization direction of $D = 43.4^\circ$, $I = -14.3^\circ$, $\alpha_{95} = 3.3^\circ$, corresponding to a paleomagnetic pole of Plat. = 13.8°, Plong. = 166.4°, $A_{95} = 2.4^\circ$.

Rock magnetic and scanning electron microscopy (SEM) studies revealed that the main carrier of the magnetization **A** is medium sized primary titanomagnetite, with varying Ti-content. This component is interpreted to be of primary origin and it provides a new well dated (1458 +4/−3 Ma) pole for Baltica. This result supports the idea that Baltica and Laurentia drifted together during the interval from 1.76 to 1.26 Ga forming the core of the supercontinent Hudsonland (Columbia, Nuna).
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1. Introduction

The Fennoscandian Shield can be divided into a number of tectonomagmatic blocks, where the oldest parts are in the northeast. Since Archaean times the amalgamation of the once separated blocks has played an important role when the Fennoscandian shield was formed. The amalgamation took place during successive orogenic events, such as the Lopian (2.9–2.6 Ga), the Svecofennian (2.0–1.75 Ga) and the Gothian (1.75–1.55 Ga), respectively. After its final formation during subsequent accretional events, the continental crust of Fennoscandia underwent major reworking during the Sveconorwegian–Grenvillian (ca. 1.2–0.9 Ga) and Caledonian orogenies (ca.0.5–0.4 Ga) ago (Gorbatschev and Bogdanova, 1993).

In the past, both continents (e.g. Baltica) and shields (e.g. Fennoscandia) have comprised various supercontinents. The amalgamation of ancient continents into supercontinents and the following break-up of them have had major impacts on geology and environment: diversification of life, unique Paleo- to Mesoproterozoic climatic conditions including several glaciations, global changes in ocean chemistry, and long-lived mantle convection patterns giving rise to plumes and large igneous provinces (e.g. Condie, 2000, 2004; Dalziel, 1994; Karlström et al., 2001; Meert, 2002; Pesonen

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et al., 2003; Rogers and Santosh, 2002; Zhao et al., 2003, 2004 and references therein). There are geological and paleomagnetic evidences that since the Mesoproterozoic three supercontinent assemblies (Rodinia ca. 1050–750 Ma, Gondwana ca. 550–400 Ma and Pangea ca. 350–165 Ma) have existed (Pesonen et al., 2003 and references therein). Evidences for the pre-Rodinia supercontinents have become more controversial as the age of the geological formations increase, due to paucity of paleomagnetic data and inherent dating problems. However, most cratonic blocks show evidences of either collisional or accretional events between 2.1 and 1.8 Ga, which has led many researchers to propose that a Mesoproterozoic supercontinent Hudsonland (also known as Columbia or Nuna) did exist in the Early Proterozoic (e.g. Condie, 2004; Meert, 2002; Pesonen et al., 2003; Rogers and Santosh, 2002; Zhao et al., 2003, 2004 and references therein). The temporal and compositional overlap between the anorogenic magmatisms in west Baltica and in east Laurentia support the model of a Mesoproterozoic supercontinent, which included Baltica and Laurentia (e.g. Åhäll and Connelly, 1998). These anorogenic events in both continents took place during the time interval of 1.7–1.3 Ga. Moreover, the 1.6–1.5 Ga is a period of tectonic stability that reflects cratonization of the newly formed lithosphere (Karlström et al., 2001). During the periods 1.7–1.6 and 1.55–1.35 Ga intracratonic bimodal a-type magmatism took place; these involved granites and associated anorthosites and related accretional tectonism (e.g. Karlström et al., 2001; Åhäll and Connelly, 1998).

Several variations of supercontinent Hudsonland have been presented, which differ from Rodinia. For example, Pesonen et al. (2003) have suggested that Laurentia and Baltica were connected for ca. 600 Ma (from 1.83 to 1.27 Ga) forming, together with Amazonia and perhaps with some other continents, like Australia, the core of the supercontinent Hudsonland. Several other authors have suggested a variety of configurations and lifecycles of this supercontinent (e.g. Karlström et al., 2001; Meert, 2002; Zhao et al., 2004 and references therein).

In order to shed light into the assembly and life time of Hudsonland we have carried out a paleomagnetic study on Valaam monzodioritic sill in the eastern part of the Fennoscandian Shield, Russian Karelia. The opportunity is excellent since the U-Pb dating of the Valaam sill yield ages of 1459 ± 3 and 1457 ± 2 Ma (Rämö et al., 2001, 2005), which are nearly coeval with St. Francois Mountain (Mnt) intrusion (1476 ± 16 Ma), Michicamau intrusion (1460 ± 5 Ma), Harp Lake Complex (1450 ± 5 Ma) and the Belt Supergroup formations (1400–1470 Ma) in Laurentia.

This compatibility provides a unique case for testing the proposed long lasting (1.8–1.2 Ga) unity of Baltica and Laurentia.

The Valaam sill provides important material for a paleomagnetic study for two reasons. Firstly, its magnetization is stable, which was by a preliminary study by Pesonen (1998) and appears to be distinct of Jotnian diabases (1.26 Ga) and Subjotnian units (1.67–1.5 Ga). Secondly, there is a lack of well dated poles between 1.6 and 1.3 Ga for Baltica (Buchan et al., 2000; Pesonen et al., 2003). However, there are two problems concerning the paleomagnetic study of the Valaam sill. First the sill is in the middle of the Lake Ladoga forming an island (Valaam) with no other older (or younger) rocks to carry out baked contact test. Second, the sill is uniformly almost horizontal (dip ca. 0.2°) ruling out the fold test possibility to prove the primary nature for natural remanent magnetization. These problems can be partially overcome by comparing the Valaam results with those from somewhat older rocks in the north of Lake Ladoga (Sortavala-Jänisjärvi area), or with known Neoproterozoic to Phanerozoic overprints (Lubnina et al., 2005; Mertanen, 2005; Salminen et al., 2006; Shcherbakova et al., 2006). The rock magnetic and scanning electron microscope (SEM) studies are also used to give further support for the primary origin of the characteristic remanent magnetization (ChRM) in Valaam sill.

2. Geological background and the sampling

Valaam island is located in the middle of the Lake Ladoga, which belongs to Russian Karelia (Fig. 1). The northeastern part of the Fennoscandian Shield is composed of Archaean rocks, which extend to ca. 100 km north from Lake Ladoga. The central part of the Shield is composed of Paleoproterozoic rocks (Gorbatschev and Bogdanova, 1993; Vaasjoki et al., 2005). The Svecofennian orogeny (ca. 1.9–1.8 Ga ago) was the main modifier of the Archaean and Early Proterozoic crust of Fennoscandian shield. The latest major crustal increment of the Shield was the emplacement of mid-Proterozoic (ca. 1.67–1.54 Ga) rapakivi granites and associate mafic bodies and tholeiitic dyke swarms (Rämö et al., 2001, 2005). In Karelia such rocks outcrop along northeast and east coast of Lake Ladoga and in the Valaam island. Three phases of magmatic activity are distinguished in the north Ladoga region. During the first two phases, basaltic lavas were erupted. In the third phase, the monzodioritic Valaam sill, which includes syenitic dikes, intruded. Its estimated thickness is 130–150 m (Amantov et al., 1996). The U-Pb baddeleyite ages of 1459 ± 3 and 1457 ± 2 Ma for Valaam sill indicate that, at least

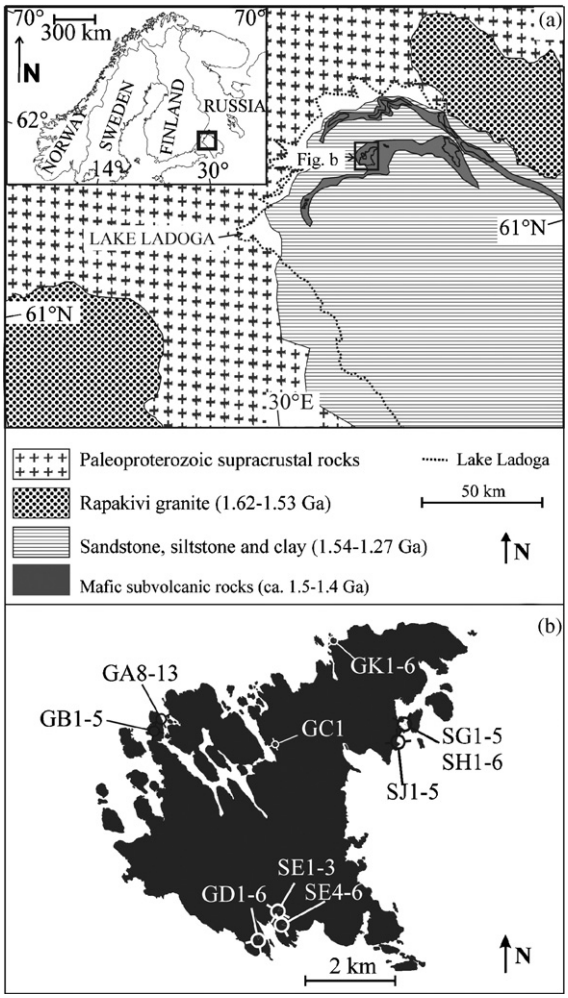


Fig. 1. The insert shows the study area (rectangle), Lake Ladoga in Russian Karelia. (a) Simplified geological map of the Lake Ladoga area (after Koistinen et al., 2001). The Valaam Island is enclosed by a rectangle. (b) Sampling sites of the Mesoproterozoic sill of the Valaam Island, where G denotes monzodioritic and S syenitic samples, respectively.

in the Lake Ladoga region, formation of the sill commenced 70–100 millions years after the emplacement of the Ladoga-Salmi rapakivi granite plutons of ca. 1560–1530 Ma in age (Rämö et al., 2001, 2005). This mid-Proterozoic (1.67–1.46 Ga) anorogenic magmatic activity formed the province that extends from south Finland to Russian Karelia in the east, to central Sweden in the west (e.g. Tuna dykes with an age of 1.46 Ga) and to Baltic countries in the south (Söderlund et al., 2005; Rämö and Haapala, 2005).

The Valaam island is a host of the Valaam Monastery, which somewhat restricts geological sampling. Excellent exposures, however, do exist all along the coast as well as in the centre of the island. Samplings for paleomagnetic studies were carried out during the summers of 2003 (western part of the island) and 2004 (eastern part of the island); sampling sites are shown in Fig. 1b. In total, 46 oriented hand samples were collected; 24 of them were greyer monzodioritic (samples G), and 22 were reddish syenitic (samples S) taken from dikes within the monzodioritic sill. Orientation was done with a magnetic compass and local declination (9°35′E; IGRF2000) was corrected.

3. Laboratory measurements and data processing

The measurements were carried out at the Solid Earth Geophysics Laboratory of the University of Helsinki. Petrophysical properties (susceptibility, intensity of NRM, and density) were measured on each specimen. Paleomagnetic measurements and alternating field (AF) demagnetizations were done using the 2 G-DC SQUID magnetometer (maximum field up to 160 mT) and AGICO-LDA-3 AF demagnetizer (maximum field up to 100 mT). A few stepwise thermal demagnetizations (temperatures up to 680 °C) were done using the Schonstedt TSD-1 furnace. During the thermal cleaning the susceptibility was monitored after each step in order to detect mineralogical changes due to heating.

Scanning electron microscope (SEM; JEOL JSM-5900LV) analysis was performed in collaboration with the Geological Survey of Finland (GSF) in order to identify the nature of the magnetic carriers. Additionally, high temperature thermomagnetic analysis (susceptibility versus temperature) for mineral identification for one sample from each site was conducted using the AGICO-KLY-3S Kappabridge. In order to avoid the oxidation during the heating the specimens were heated in argon gas from room temperature up to 700 °C and cooled back to room temperature. For determination of domain states of the magnetic carriers, hysteresis properties were measured using a Princeton Measurement Corporation's MicroMag^{TM}3900 model Vibrating Sample Magnetometer (VSM).

The anisotropy of the weak field magnetic susceptibility (AMS) was measured for one specimen from each hand sample in order to define the magnetic fabric of the sill at different sites (e.g. Dragoni et al., 1997; Elming and Mattsson, 2001; Hrouda, 1982).

Components were isolated by a least squares method (Kirschvink, 1980; Leino, 1991). Mean remanence directions of various components were calculated using Fisher (1953) statistics. The paleoreconstructions were plotted with the GMAP programme (Torsvik and Smethurst, 1999).

Table 1
Site-mean petrophysical properties of the Valaam sill

Site	*B*/*N*/*n*	*D* (kg m^{-3})	*k* (10^{-6} SI)	J_r (mA m^{-1})	*Q*
Monzodiorite					
GA	1/6*/137	2858 ± 7	46378 ± 8589	1832 ± 380	1.0 ± 0.2
GB	1/5*/73	2846 ± 27	42281 ± 6169	3084 ± 577	1.9 ± 0.4
GC	1/1*/18	2796	31493	1904	1.5
GD	1/6*/113	2787 ± 17	34457 ± 10222	1556 ± 1066	1.0 ± 0.4
GK	1/6*/28	2859 ± 24	40191 ± 6769	2393 ± 716	1.5 ± 0.3
Mean	5/24*/369	2829 ± 31	38960 ± 5359	2154 ± 538	1.4 ± 0.3
Syenitie					
SG	1/5*/46	2642 ± 60	19874 ± 13956	1366 ± 1265	1.5 ± 0.3
SH	1/6*/50	2606 ± 38	5103 ± 3899	272 ± 195	1.6 ± 0.2
SJ	1/5*/39	2666 ± 10	12745 ± 5490	863 ± 432	1.7 ± 0.6
SE	1/6*/90	2681 ± 16	15038 ± 3137	892 ± 344	1.4 ± 0.3
Mean	4/22*/225	2649 ± 33	13190 ± 6156	848 ± 448	1.5 ± 0.2

B/*N*/*n* is the number of sites/samples/specimens; (*) denotes the statistical level used for mean calculations; *D* density; *k* is susceptibility; J_r the intensity of the natural remanent magnetization; *Q* the Koenigsberger's ratio calculated for a field of 50 μT; the standard deviations are for each site and for the mean value.

4. Petrophysical results

Petrophysical properties (density, susceptibility, NRM intensity and *Q*-value) for the monzodioritic and syenitic samples are summarized in Table 1. The bivariate diagrams of density versus susceptibility and NRM versus susceptibility are shown in Fig. 2. Since, the chemical and mineralogical compositions of the syenitic and monzodioritic samples are distinct, their bulk petrophysical properties differ as clearly seen in Fig. 2 (Table 1). For example, the monzodiorites have distinctly higher density (Fig. 2 a) as expected due to higher content of heavier (e.g. iron bearing) minerals, such as hornblende ($Ca_2(Fe, Mg)_4Al(Si_7Al)O_{22}(OH, F)_2$), pyroxene ($(Ca, Fe, Mg)_2Si_2O_6$) and biotite ($K(Mg, Fe)_3AlSi_3O_{10})(OH, F)_2$, than in the syenites. Similarly, the monzodiorites have somewhat higher remanence intensities than the syenites; however, there is no marked difference in the *Q*-values. However, within the two groups the variation of petrophysical properties is small indicating that the magmas were quite homogenous.

5. Paleomagnetic results

Both monzodioritic and syenitic samples show two remanent magnetization components: a low coercivity component, likely of viscous origin, which is interpreted to be related to the present earth field (PEF), and a high unblocking/coercivity component (**A**) with a shallow northeast upward inclination ($D = 43.4°$, $I = -4.3°$, $\alpha = 3.3°$) (Fig. 5). Occasionally, a third component with a shallow northeast downward inclination is seen, but this

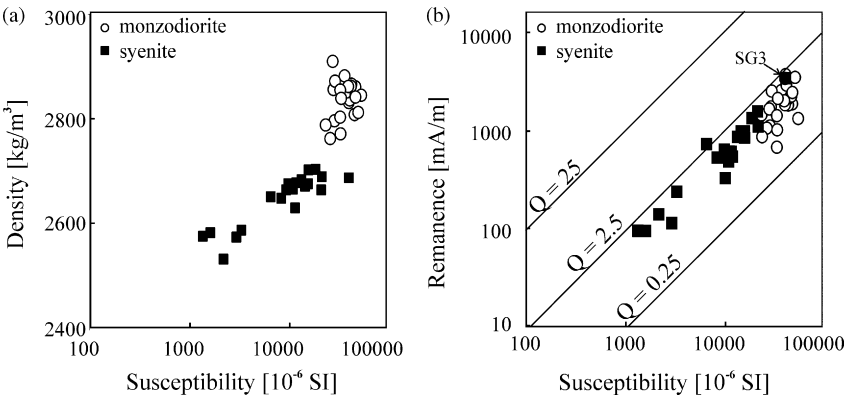


Fig. 2. (a) Density-susceptibility (semilogarithmic scale) and (b) remanence-susceptibility (logarithmic scale) diagrams for samples of the Valaam sill. Koenisgsberger's *Q* ratios (calculated with the field value of 50 μT) are shown as straight lines in (b).

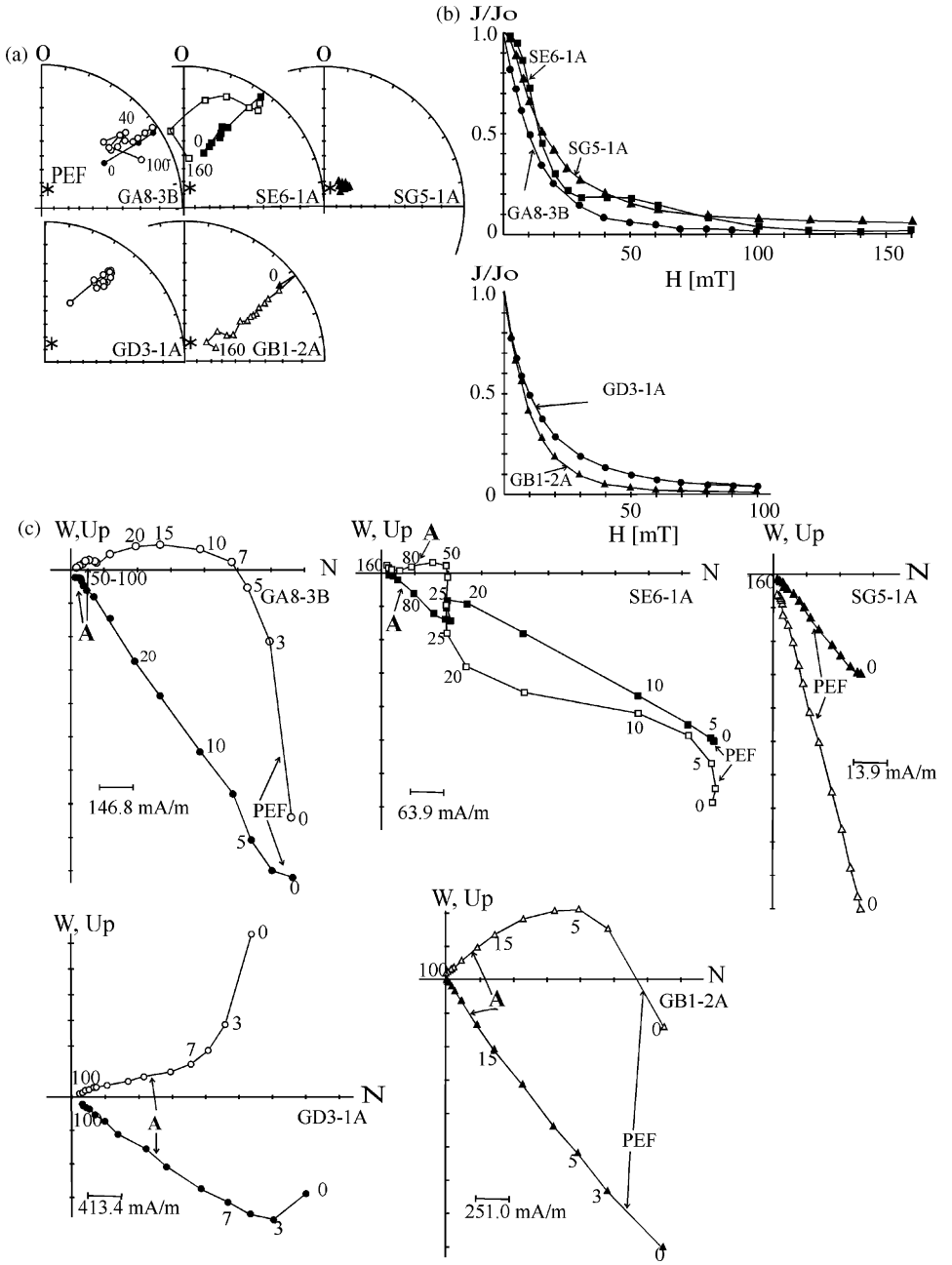


Fig. 3. Examples of AF demagnetization behaviour of specimens of the sill. Specimen GA8-3B represents the most typical behaviour of the samples. (a) Stereographic projections of directional data upon demagnetization, where PEF denotes the direction of the present Earth's magnetic field, (b) relative intensity decay, (c) orthogonal (Zijderveld) demagnetization diagrams. Open (closed) symbol denotes vertical (horizontal) plane. Number at each demagnetization step denotes the field value (mT). G (S) at the beginning of the specimen's name denotes monzodioritic (syenitic) sample.

may be an artefact. Examples of AF and thermal demagnetizations of the samples are given in Figs. 3 and 4. The site-mean remanence directions for component **A** are well clustered (Fig. 5). The mean directions and the corresponding paleomagnetic poles are averaged for all the sites using Fisher statistics, and are shown in Table 2.

5.1. Alternating field demagnetization

Approximately three to seven specimens from every sample were AF demagnetized. However, in case of monzodioritic samples GK5 and GK6, due to small size of the samples, only one specimen of each were

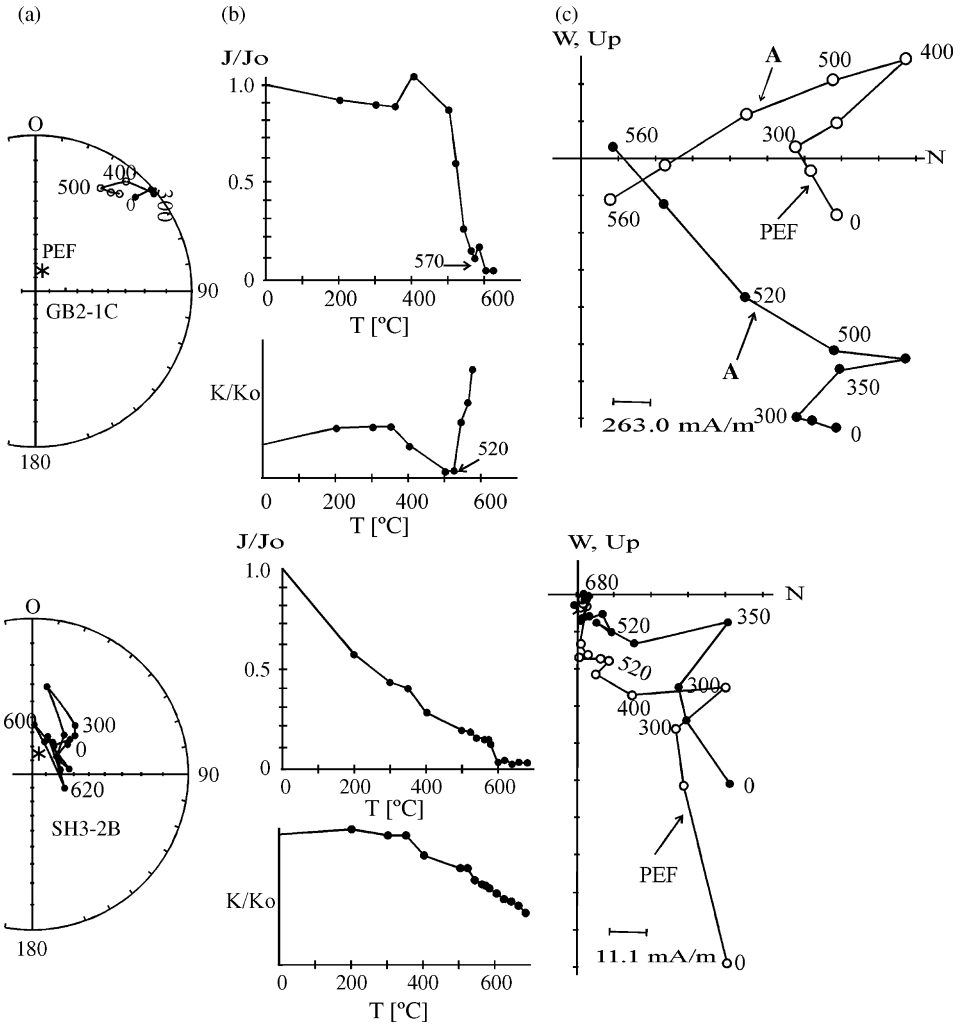


Fig. 4. Examples of thermal demagnetization behaviour of specimens of the Valaam sill. (a) stereographic projections, (b) relative intensity decay (upper) and behaviour of susceptibility upon heating (lower), (c) orthogonal (Zijderveld) demagnetization diagrams. Number at each demagnetization step denotes the temperature (°C). Other symbols as in Fig. 3.

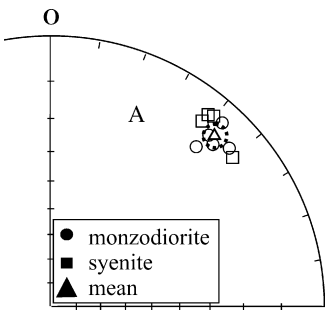


Fig. 5. Site-mean directions of component **A** for the Valaam monzodioritic (circle) and syenitic (square) samples. Component's mean direction is shown as a triangle with corresponding 95% confidence circle (dotted line). Open symbols denote upward directions.

treated. Most of the samples show a steep downward direction that is removed in the first steps of the AF demagnetization (0–2.5/5 mT). Because it is aligned approximately along the PEF direction at the sampling site, it is most likely a viscous remanent magnetization (VRM) of recent origin.

The most typical natural remanent magnetization (NRM) resultant vector moves from an intermediate northeast downward direction to a shallow upward northeast direction in the course of AF treatment (in 24 samples; for example, specimens GA8-3B and SE6-1A; Fig. 3). Some specimens show a stable single component, with an intermediate upward northeast direction (specimen GD3-1A; Fig. 3) or a moving resultant from a steep northeast downward direction to shallow northeast upward direction (specimen GB1-2A; Fig. 3). This high

Table 2
Mean paleomagnetic data for the monzodiorite and syenite (Lat. 61°23′N, Long. 30°55′E)

SITES	*B*/*N*/*n*	*D* (°)	*I* (°)	*k*	α95 (°)	Plat (N°)	Plong (E°)	dp (°)	dm (°)	A_{95} (°)	*S* (°)
High coercivity (*A*)											
GA	1/6*/36	42.7	−15.9	144.2	5.6	12.7	166.6	3.0	5.8	4.2	6.8
GB	1/4*/17	48.0	−14.1	28.0	17.7	12.1	161.9	9.2	18.1	12.9	15.3
GC	1/1*/5	44.8	−17.3			11.6	165.6				
GD	1/6*/30	42.3	−21.6	19.9	15.4	11.0	168.7	8.6	16.2	11.8	18.2
GK3-5	1/3*/9	43.0	−9.6	90.6	13.0	16.2	165.9	6.7	13.2	9.4	8.5
SE1, SE3-6	1/5*/23	39.5	−13.8	37.2	12.7	15.4	170.1	6.6	13.0	9.3	13.3
SG1-3	1/3*/11	40.7	−9.8	118.3	11.4	16.9	168.2	5.8	11.5	8.2	7.4
SH1, 5, 6	1/3*/9	50.0	−15.3	13.6	34.8	11.0	160.4	18.3	35.7	25.6	22.0
SJ	1/5*/14	39.6	−10.8	59.3	10.0	16.8	169.5	5.1	10.2	7.2	10.5
Mean *A*	9*/36/154	43.4	−14.3	239.1	3.3	13.8 (N) −13.8 (R)	166.4 (N) 346.6 (R)	1.7	3.4	2.4	12.8

G (S) in sample name denotes monzodioritic (syenitic); *B*/*N*/*n* = number of sites/samples/specimens, where (*) denotes the statistical level used for mean calculations; *D*, *I* = declination, inclination of the remanent magnetization component, respectively; *k* = Fisher's (1953) precision parameter, α_{95} = radius of circle of 95% confidence; Plat, Plong = latitude and longitude for the paleomagnetic pole (N, R = normal, reversed polarity); dp and dm = the semi axes of the oval of 95% confidence of the pole; A_{95} = the radius of the circle of 95% confidence of the pole; *S* = estimated angular dispersion.

coercivity ChRM component (**A**) has a shallow northeast negative (upward) inclination. It was typically isolated in the range of 30–100/160 mT. In most samples the vector trajectory in the Zijderveld (1964) diagrams (Fig. 3 c) goes towards the origin indicating that there are no other higher coercivity components present.

Many of the samples show shallow downward direction that is removed in low to intermediate AF fields (5–30 mT). However, due to the fact that it is expressed with curved vector plots (Zijderveld plots) it is probably an artefact (overlapping coercivity spectras of PEF and component **A**). However, the other paleomagnetic studies from Fennoscandia (e.g. Mertanen and Pesonen, 1994 and references therein; Mertanen et al., 2006) have shown the presence of similar secondary magnetization as an overprint.

5.2. Thermal demagnetization

Thermal demagnetization was carried out for 17 specimens. Results are similar to those of AF demagnetization, revealing also two remanence components: PEF and **A**. Samples showing dominance of PEF (GD3, GK1, SG5, SH3 and SH4) lost half of their initial magnetization in temperature range 200–350 °C. In contrast, samples showing also **A**-component lost half of their initial magnetization in temperature around 500 °C. Most of the samples lose their measurable magnetization around 520–600 °C. Besides, in many thermally demagnetized samples a new magnetic mineral emerged at temperatures above 520–560 °C. This was observed by monitoring the bulk susceptibility, which was increasing after 520–560 °C (for example specimen GB2-1C; Fig. 4). Therefore, in many samples it was not possible to isolate any stable component beyond 520–560 °C.

Laboratory unblocking temperatures of magnetization for **A**-component range between 350 and 600 °C, mostly around 550 °C, indicating that titanomagnetite is the carrier of remanent magnetization of the **A** component (Fig. 4). This is confirmed with Curie-point determinations (see Section 6), which generally yield Curie points ca. 535–560 °C. A change at about 330 °C in both the thermal demagnetization decay and in susceptibility versus temperature curves may indicate that a small amount of higher Ti-content titanomagnetite or pyrrhotite is present.

6. Remanence carriers of the samples

We performed thermomagnetic measurements (Curie-point analysis), hysteresis measurements and SEM analysis in order to identify the remanence carriers of the samples. Table 3 summarizes the hysteresis and thermomagnetic results of the different samples.

Thermomagnetic measurements were carried out for one sample for each of the nine sampling sites. Curie points were determined using the maximum in the second derivatives of the thermomagnetic curves, since it occurs at the point of maximum curvature in the thermomagnetic curve and it is a reasonable estimate of the Curie point (e.g. Tauxe, 2002). Each Curie temperature shown in the thermomagnetic curve characterizes a ferromagnetic mineral or phase in the rock. The pos-

Table 3
Hysteresis properties and Curie points for the sill samples

Sample	M_r (mA m^2/kg)	M_s (mA m^2/kg)	H_{cr} (mT)	H_c (mT)	M_r/M_s	H_{cr}/H_c	T_C (°)
Monzodioritie							
GA8	66.7	1131.0	24.8	5.6	0.06	4.41	
GA9	10.3	84.9	23.68	9.5	0.11	2.49	535
GA11	338.8	2441.0	28.0	12.7	0.15	2.21	
GA12	141.3	3169.0	16.7	4.3	0.05	3.89	
GB1	606.2	8241.0	22.5	6.3	0.07	3.56	
GB2	348.8	2154.0	27.9	12.3	0.18	2.28	535
GB3	109.3	1360.0	17.8	6.9	0.08	2.58	
GD1	26.4	215.7	23.6	11.1	0.12	2.12	550
GD3	60.3	646.0	18.3	6.3	0.11	2.46	
GD4	105.9	994.6	24.4	9.9	0.11	2.46	
GD5	240.3	2465.0	23.8	8.2	0.10	2.92	
GD6	406.5	2629.0	32.0	14.6	0.16	2.20	
GC1	46.0	319.4	23.7	10.2	0.14	2.31	
GK2	81.0	1311.0	25.7	6.3	0.06	4.09	
GK3	439.7	2273.0	29.1	13.6	0.19	2.15	535
GK6	85.6	565.2	36.1	15.5	0.15	2.33	
Mean	194.6 ± 179.4	1875.0 ± 1953.6	24.9 ± 5.1	9.6 ± 35	0.12 ± 0.04	2.78 ± 0.76	
Syenite							
SE1	79.0	522.6	21.8	11.4	0.15	1.91	
SE3	182.7	2121.0	25.5	6.9	0.13	2.46	565
SE4	62.8	440.6	32.9	12.2	0.15	2.71	
SG1	262.2	2013.0	26.9	11.2	0.13	2.40	
SG3	183.9	2112.0	25.5	6.9	0.09	3.70	325, 560
SG4	3.1	28.3	40.5	9.8	0.11	4.15	
SG5	6.6	50.1	34.5	17.1	0.13	2.02	340, 560
SH4	9.6	88.3	29.3	8.9	0.11	3.31	555
SH5	6.4	19.3	37.5	16.2	0.33	2.31	
SJ3	0.5	28.3	15.0	2.6	0.02	5.82	355, 560
SJ5	227.8	778.0	43.5	24.5	0.29	1.77	
Mean	93.1 ± 101.4	745.6 ± 893.4	30.3 ± 8.5	11.6 ± 6.0	0.15 ± 0.09	2.96 ± 1.21	

M_r = saturation remanence; M_s = saturation magnetization; H_{cr} = coercivity of remanence; H_c = coercive force and T_C = Curie point(s) observed from the heating curve.

sible differences in heating and cooling curves indicate chemical changes in minerals during the thermal treatments. Samples GD1, GK3 and SE3 (Fig. 6) increase in susceptibility in the heating curve just before the Curie temperature (a Hopkinson peak). This is characteristic behaviour of single domain (SD) and pseudo single-domain (PSD) magnetite grains (Dunlop, 1974). Heating and cooling curves show Curie temperatures from 535 to 560 °C indicating presence titanomagnetite, with minor Ti-content. The differences between heating and cooling curves of these samples suggest that during the heating some mineral changes occur (e.g. sample GA9). This was also observed during the thermal demagnetization, which showed the growth of new minerals after 520–560 °C. Samples SG3 (Fig. 6d) and SJ3 (Table 3) behave similarly to the other samples, but there is also a decline in the susceptibility in the temperature range of 325–355 °C, which may be due to either more Ti-rich titanomagnetite, ilmenohematite or the presence of pyrrhotite. The shape of the thermomagnetic curve favours the Ti-rich titanomagnetite or ilmenohematite (Fig. 6d).

As a summary, the Curie point of the monzodioritic samples is somewhat lower (ca. 535 °C) than the Curie point of the syenitic samples (ca. 550 °C). In both cases the main ferromagnetic mineral seems to be titanomagnetite. Some syenitic samples show also evidences of Ti-rich titanomagnetite or ilmenohematite with Curie points ≤ 350 °C.

Because the stability of the NRM depends on the grain sizes of the remanence carriers, we measured the hysteresis properties of the samples in order to define the magnetic domain types. Fig. 7 depicts the hysteresis loops for four specimens. All the samples were

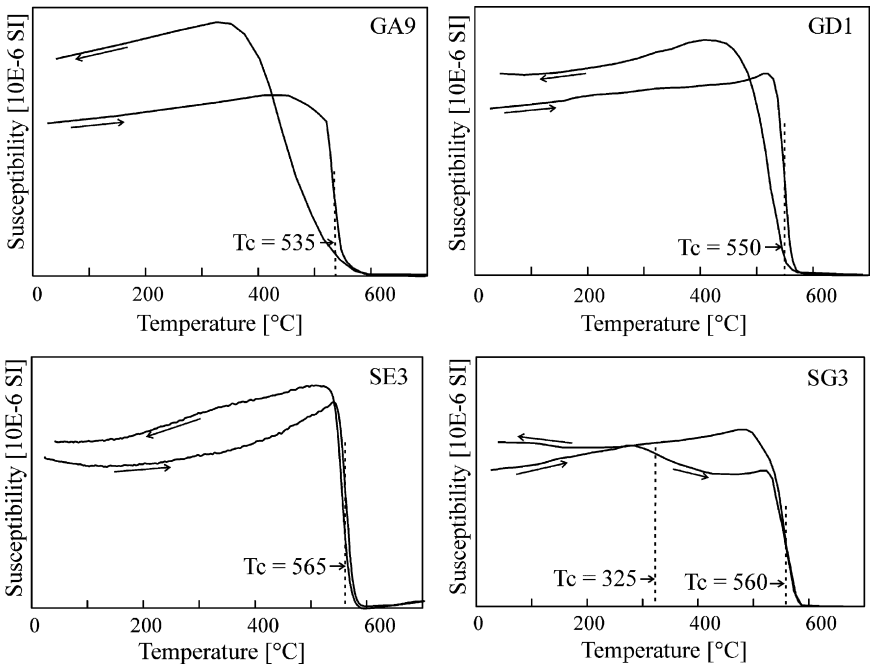


Fig. 6. Thermomagnetic curves (susceptibility vs. temperature) for specimens of the Valaam sill. T_C denotes the Curie temperatures as defined by the second derivative method (e.g. Tauxe, 2002).

saturated below 0.5 T, which also supports the presence of (titano)magnetite. According to Dunlop (1983) (titano)magnetite saturates between 0.3 and 0.5 T. In view of the fact that titanomagnetite is the dominant magnetic phase, we may interpret the hysteresis ratios using the Day plot (Day et al., 1977; Fig. 8; Table 3) in terms of domain states. In our analysis we use the re-appraisal of the Day plot after Dunlop (2002). In general; it appears that samples fall close to the boundary between multidomain (MD) and single-domain (SD) mixtures in the area of pseudo single domain (PSD). There are no differences in domain states between syenites and monzodiorites; however, the syenites seem more scattered. AF and thermal demagnetization data indicate the presence of a PEF component superimposed on a stable characteristic component, which also suggests the mixture of grain sizes of MD–SD range.

SEM studies of two monzodioritic and two syenitic samples established the presence of nearly pure magnetite grains, many unaltered titanomagnetite grains (up to few hundreds of micrometer) and many altered titanomagnetite grains, where secondary ilmenohematite formed (Fig. 7). The SEM analysis clearly shows that Ti-content in titanomagnetite varies (from 0 wt% up to 15 wt%) and some of the titanomagnetite has partly oxidized to ilmenohematite (Fig. 7). Such an oxidation process is normally interpreted as a consequence of low temperature hydrothermal alteration (Dunlop and Özdemir, 1997).

7. Anisotropy of susceptibility

The anisotropy of susceptibility (AMS) measurements were carried out in order to study the magnetic fabric of the sill, since AMS principal directions can record flow-directions from magma (Dragoni et al., 1997; Elming and Mattsson, 2001). For reliable paleomagnetic studies the obtained remanence direction should reflect accurately the Earth's ambient magnetic field direction. However, the remanence direction may deviate from the ambient field direction due to magnetic anisotropy. It has been suggested that the NRM will not be significantly altered if the ratio between the maximum and minimum suceptibilities (degree of anisotropy) is lower than 5% (Hrouda, 1982).

The degree of anisotropy (P) in both monzodioritic and syenitic samples varies between 1 and 6% (average 2.7%; Table 4; Fig. 9a). As this is so low, it has not significantly deflected the characteristic remanence direction. Monzodiorites show slightly higher P and anisotropy shape parameter values (T) than syenites. For most rocks the direction of the maximum susceptibility axis corresponds to the mineral lineation (L) (Hrouda, 1982), but in the Valaam samples there is

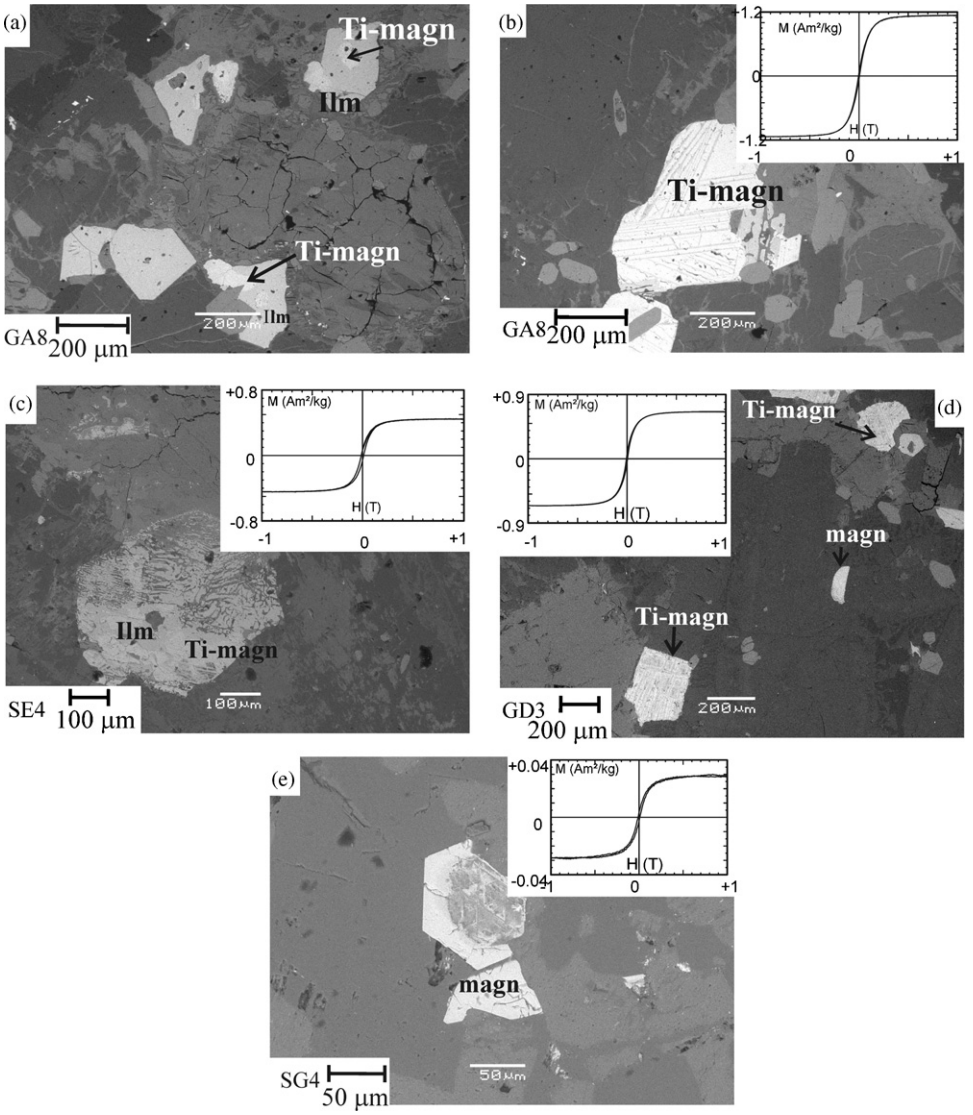


Fig. 7. SEM (JEOL JSM-5900LV) figures of magnetic minerals and hysteresis loops for Valaam monzodioritic and syenitic samples. (a) monzodioritic sample GA8 showing altered Ti-magnetite, (b) same monzodioritic sample GA8 showing unaltered Ti-magnetite and corresponding hysteresis loop, (c) syenitic sample SE4 showing altered Ti-magnetite and corresponding hysteresis loop, (d) monzodioritic sample GD3 showing unaltered Ti-magnetite and magnetite grains and corresponding hysteresis loop, (e) syenitic sample SG4 showing unaltered magnetite and corresponding hysteresis loop.

no preferred direction of magnetic lineation (Fig. 9b). Usually this is caused by the limited number of samples (Dragoni et al., 1997; Elming and Mattsson, 2001), but in this case (46 samples) this most likely indicates a true absence of preferred direction of lineation, i.e. the samples have been collected from the central, maybe strain free part, of the sometimes 150 m thick sill body where no preferred direction exists. Dragoni et al. (1997) pointed out that the centre of the sill body is strain free and no preferred orientation can be defined.

8. Discussion

A viscous component related to PEF was isolated in almost all samples. In most of the samples this component was erased during the first steps of demagnetization; however some syenitic samples show only single hard overprint close to the PEF direction. The characteristic remanent magnetization (component **A**) is well defined and similar in both syenitic and monzodioritic samples, indicating that the samples recorded the ambient field simultaneously (Fig. 5). According to thermomagnetic,

Table 4
Parameters of the anisotropy of susceptibility for sill samples

Specimen	*P*	*L*	*F*	Specimen	*P*	*L*	*F*
GA8-3B	1.038	1.013	1.024	SE1-5D	1.012	1.003	1.009
GA8-4A	1.039	1.005	1.033	SE1-7B	1.015	1.005	1.009
GA9-6C	1.048	1.008	1.039	SE2-5A	1.020	1.008	1.012
GA10-5A	1.042	1.007	1.035	SE2-7B	1.018	1.011	1.007
GA11-4A	1.038	1.006	1.031	SE3-2C	1.040	1.017	1.022
GA12-3C	1.035	1.007	1.028	SE3-3C	1.018	1.009	1.009
GA13-1C	1.045	1.006	1.039	SE4-2B	1.046	1.010	1.036
GA13-2A	1.028	1.006	1.022	SE5-2E	1.025	1.004	1.021
GB1-1A	1.033	1.007	1.026	SE6-2A	1.020	1.011	1.009
GB2-1B*	1.019	1.008	1.011	SG1-1E*	1.023	1.008	1.015
GB3-3B*	1.057	1.008	1.049	SG2-1B*	1.017	1.007	1.010
GB4-1B	1.027	1.004	1.023	SG3-1B*	1.034	1.025	1.008
GB5-1A*	1.026	1.006	1.020	SG4-2A*	1.019	1.015	1.005
GC1-3D	1.015	1.004	1.011	SG5-1A*	1.005	1.003	1.002
GC1-4D	1.011	1.008	1.003	SH1-2D	1.037	1.004	1.033
GD1-6D	1.033	1.010	1.024	SH2-1A*	1.011	1.009	1.001
GD2-2B	1.045	1.006	1.039	SH3-1A*	1.020	1.013	1.007
GD3-3C	1.023	1.005	1.018	SH4-2A*	1.015	1.009	1.006
GD4-1B	1.022	1.013	1.010	SH5-1A	1.012	1.007	1.004
GD5-4B	1.022	1.007	1.015	SH6-2B	1.030	1.003	1.027
GD6-2D	1.027	1.010	1.017	SJ1-1B	1.020	1.006	1.013
GK1-1A	1.027	1.002	1.025	SJ2-1A	1.013	1.008	1.005
GK1-2B*	1.026	1.003	1.023	SJ2-3A	1.026	1.005	1.021
GK2-1A*	1.034	1.011	1.023	SJ3-3A*	1.013	1.009	1.004
GK3-3A*	1.031	1.003	1.028	SJ4-2A*	1.022	1.011	1.011
GK4-1A	1.032	1.004	1.028	SJ4-3A	1.016	1.003	1.013
GK5-1A	1.055	1.008	1.047	SJ5-1A*	1.014	1.007	1.006
Mean monzodiorite	1.033	1.007	1.026	Mean syenite	1.021	1.009	1.012

P is the ratio between maximum and minimum susceptibility; *L* (lineation) is the direction of maximum susceptibility; *F* (foliation) is the plane perpendicular to the minimum magnetic susceptibility; symbol (*) marks the specimen where alternating field demagnetization performed before anisotropy of susceptibility (AMS) measurement.

high field and SEM analyses this component is carried by PSD or MD titanomagnetite (varying Ti-content) (Fig. 7). Additionally, an intermediate component was also isolated. However, the trajectories of this component in Zijderveld plots are seldom straight lines, which may be due to the fact that this component is not a true component, but combination of component **A** and PEF, which have overlapping coercivity spectras.

8.1. Paleomagnetic poles

The site-mean paleomagnetic pole for **A** (Plat = 13.8°N, Plong = 166.4°E, $A_{95} = 2.4°$) is calculated for the Valaam sill data from the magnetization data of each sampling site (Table 2). We were not able to carry out any field tests to prove a primary origin of the **A** component, because there where no contacts with the host rock for a baked contact test. The regional tilting of the shield is, at maximum, about 0.2° (Laitakari et al., 1996), which does not affect to the results. However the

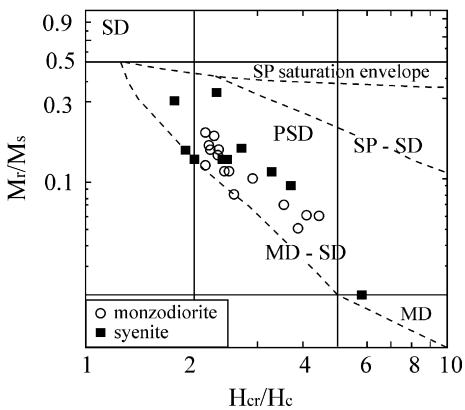


Fig. 8. Measured hysteresis ratios for the Valaam sill plotted on the Day plot (Day et al., 1977; Dunlop, 2002). M_r, M_s saturation remanence, saturation magnetization; H_c coercive force; H_{cr} coercivity of remanence; SD single domain; PSD pseudo single domain; MD multi domain.

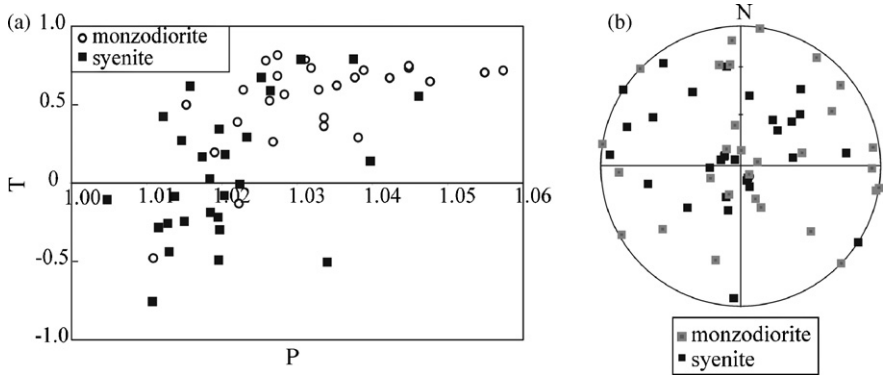


Fig. 9. (a) The anisotropy shape parameter T (Jelínek, 1981) plotted as a function of the degree of anisotropy (P) of the Valaam sill samples, (b) equal area stereoplot of the directions of magnetic lineations (maximum axes of susceptibility) of the sill samples. The directions are plotted on lower hemisphere.

A pole is clearly distinct from the poles of Svecofennian age (ca. 1.8 Ga) and from the pole (Plat = 57.6°N, Plong = 268.8°E, A_{95} = 13.6°; Salminen et al., 2006) obtained for the Svecofennian target rocks (ca. 1.88 Ga) of the Jänisjärvi impact crater, which is located ca. 100 km north from Valaam island (Fig. 10). In addition, thermal, hysteresis and SEM analyses support the fact that the component **A** is primary and carried by primary titanomagnetite.

When comparing the normal polarity $\mathbf{A}_N$-pole to the Proterozoic key poles (Fig. 10; Table 5), it plots in between ca. 1.6 and 1.27 Ga key poles but off from the poles. Similarly, a normal polarity pole reported in preliminary study for Mesoproterozoic olivine dolerite

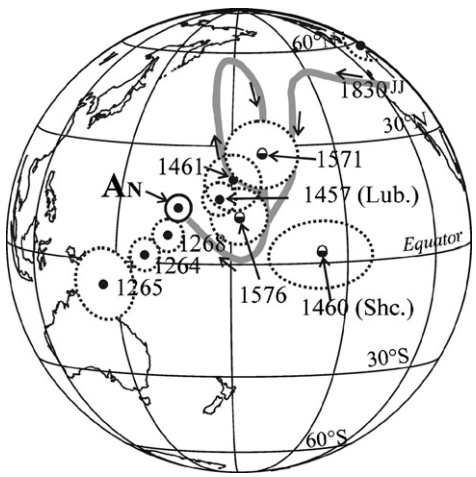


Fig. 10. Paleomagnetic pole A of this study with its 95% confidence circle plotted among paleomagnetic key poles (Elming et al., 2001) and other relevant Mesoproterozoic poles (Table 5). Lub. represents data from Lubnina et al., 2005; Sch. represents data from Shcherbakova et al., 2006. Open (closed) circle denotes reversed (normal) polarity. Half open and half closed circle denotes mixed polarity.

dyke swarms at the northern Lake Ladoga area (Lubnina et al., 2005; interpreted age 1457 ± 2 Ma). This pole is plotted in Fig. 10 and it is located only slightly off from the Valaam $\mathbf{A}_N$-pole. Moreover, a recent study by Shcherbakova et al. (2006) provided a new reversed polarity pole for Salmi (north Lake Ladoga; interpreted age 1460 Ma) volcanic and subvolcanic rocks (Fig. 10), which is ca. 20° off from other Lake Ladoga poles and also off from the apparent polar wander path (APWP). Possible reasons for this anomalous position might be: a slight difference between the magnetization ages of the sampling sites; secular variation, which is not averaged out; or unknown structural tilting. For two first reasons it would be useful to use the mean pole from these sites, which also provides a better average for the secular variation.

Using only the Valaam results, the paleomagnetic approach allows us to locate Baltica either on the southern hemisphere (7°S), or upside down on the northern hemisphere (7°N). It is important to note that this places Baltica at low latitudes at 1.46 Ga. Further support of this low latitude location for Fennoscandia during the Mesoproterozoic is for example the Satakunta red-bed sedimentation (Rämö et al., 2005).

For paleomagnetic study it is important that a paleomagnetic pole represents a sampling of the geomagnetic field over a time interval that exceeds the dominant periodicities of secular variation. Analyses of the recent geomagnetic field have shown that the dominant periodicities of secular variation are ca. 10^5 year (e.g. Merrill and McElhinny, 1983 and references therein). The angular dispersion (S) is measure for this scatter (see Table 2) and varies as a function of latitude. Calculated angular dispersion for a given data set (e.g. Valaam) has to be compared with theoretical dispersion, e.g. with the model by Merrill and McElhinny (1983) on the pale-

Table 5
Proterozoic key poles and selected Early Cambrian to Upper Riphean poles for Baltica

Rock unit	Age (Ma)	Pol	D_R	I_R	Plat (°N)	Plong (°E)	A_{95} (°)
Proterozoic key poles (Buchan et al., 2000)							
Åland quartz porphyry dykes	1576 ± 13	M	26	−22	12	182	7
Åland dolerite dykes	1571 ± 9	M	18	7	28	188	9
Tuna dykes[a]	1461–1462[b]	N	26	−5	21	180	7
Salmi Frm volcanic rocks[c]	1460	M	8	−36	6	200	11
Lake Ladoga olivine dolerite[d]	1457 ± 2	N	30	−13	16	177	4
Valaam monzodioritic sill, A pole[e]	1458 +4/−3	N	41	−11	14	166	2
Mean for Baltica	ca. 1460	M	26	−16	15	181	11
Vaasa dolerite dykes	1268 ± 13	N	45	−23	7	164	4
Satakunta dolerite dykes and sills	1264 ± 12	N	52	−27	2	158	4
Märket dolerite	1265 ± 6	N	67	−32	−6	146	9

Age is rock age, Pol is polarity (N, R, M normal, reversed, mixed), D_R, I_R refer to declination, inclination of a central reference location for each craton/continent; reference location for Baltica: 64°N, 28°E; Plat/Plong pole latitude/longitude, A_{95} the confidence circle of the pole

[a] Bylund (1985).
[b] Söderlund et al. (2005).
[c] Shcherbakova et al. (2006).
[d] Lubnina et al. (2005).
[e] This work.

olatitude of the sampling site (7°). When comparing the observed between site dispersion of site-mean paleopoles for component **A** we obtain a value of $S = 12.8°$ (see Table 2) which is consistent with the paleolatitude of 7° suggesting that the secular variation has been adequately sampled, since the observed angular dispersion of site-mean poles is consistent with that already predicted (ca. 13°).

8.2. Unity of Baltica and Laurentia during Mesoproterozoic?

Using the new 1.46 Ga paleomagnetic pole for Baltica we are able to test the existence of the proposed long lasted (about 600 Ma) unity of Baltica and Laurentia during the Mesoproterozoic (from 1.83 to 1.27 Ga). Pesonen et al. (2003) proposed that Baltica and Laurentia were joined together during this period by rotating counterclockwise. Since their research is an extensive summary from a paleomagnetic point of view; we will use it as a base for our study.

At 1.46 Ga the position of Baltica is based on the well dated paleomagnetic $\mathbf{A}_N$-pole from Valaam sill combined with poles from Salmi volcanic rocks, Lake Ladoga dolerites, and Tuna dykes (Fig. 11; Tables 5 and 6). Moreover, a recent preliminary study of Mesoproterozoic dykes in Central Sweden by Lubnina et al. (2007) provides further support to this pole position. Laurentia's position is reconstructed using the mean pole from St. Francois Mnt (1.476 Ga), combined with results from Michicamau intrusion and Harp Lake complex, with magnetization ages about 1.445 Ga (see Meert and Stuckey, 2002) and the 1.46 Ga Spokane formation from Belt Supergroup (Vitorello and Van der Voo, 1977) (Fig. 11; Table 6). Since the poles for Baltica and Laurentia are not necessary coeval and they do not have exactly the same age it is permissible that their ovals of confidence are just overlapping with the geographic pole and with each others (Fig. 11). The Euler poles used

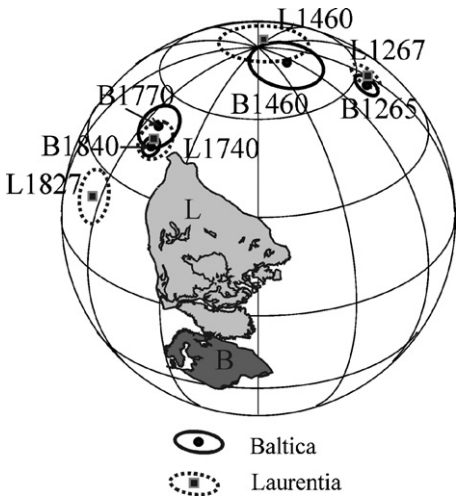


Fig. 11. Reconstruction of Baltica and Laurentia at 1.46 Ga based on mean poles for Baltica and Laurentia (Table 6; Euler pole for Baltica is Lat: 0°, Long: 102°, angle 79° and for Laurentia is Lat: 0, Long: 312, angle: −93.5). Also shown 1.83, 1.76, 1.46 and 1.27 Ga poles for Baltica and Laurentia according to this continental configuration. B (L) stands for Baltica (Laurentia). Numbers are ages of used poles in million years.

Table 6
Paleomagnetic poles used for reconstructions

Cra	Formation	D_R (°)	I_R	Plat (°)	Plong (°)	A_{95}, dp/dm (°)	Age (Ma)/method	Original references
Reconstruction at ~1.83 Ga (after Pesonen et al., 2003)								
B	Haukivesi dykes	348	40	48.0	225.0	3.0	1840/U-Pb	Neuvonen et al. (1981)
L	Sparrow dykes	159	60	12.0	291.0	8.0	1827 ± 4/U-Pb	McGlynn et al. (1974), Bostock and van Breemen (1992)
Reconstruction at ~1.77 Ga								
B	Ropruchey-Shosksha frm	350	31	42.0	221.0	7	ca. 1770	Pisarevksy and Sokolov (2001)
L	Cleaver lake dykes	197	60	19.0	263.0	6	1740+5/−4	Irving et al. (2004)
Reconstruction at ~1.5 Ga								
B	Mean for Baltica	26	−16	15	181	11	ca. 1460	This work
L	Belt super group mean (BSG)	242	−12	−18.9	207.2	5.6	1400–1470	Elston and Bressler (1980)
L	Laramie anorthosite/Sherman granite (LA)	240	17	−6.7	215.0	3.5	1429 ± 9	Harlan et al. (1994)
L	Electra Lake Gabbro (ELG)	229	−4	−21.1	221.1	3.4	1433 ± 2	Harlan and Geissman (1998)
L	Spokane Frm-Belt Supergroup (SPF)	227	9	−15.5	225.5	5.1	ca. 1460	Vitorello and Van der Voo (1977)
L	Michikamau Intr. (MI)	240	27	−1.5	217.5	3.3/6.6	1460 ± 5/U-Pb	Emslie et al. (1976)
L	Harp Lake Compl. (HL)	252	23	1.6	206.3	3.0/6.1	1450 ± 5/U-Pb(z)	Irving et al. (1977)
L	St. Francois Mtn (SFM)	234	9	−13.0	219.0	7.0	1476 ± 16	Meert and Stuckey (2002)
L	Mean (SPF, MI, HL, SFM)	239	18	−7.0	217.0	13.2	1460	Interpolated
Sb	Kuonamka dykes	52	−25	6.0	234.0	20.0	1503 ± 5/U-Pb	Ernst et al. (2000)
Au (NA)	Mt Isa block Intr.	185	49	−79.0	111.0	8.0	1550–1500	Tanaka and Idnurm (1994)
Au (SA)	Gawler Range Volcanics	34	−47	60.4	230.0	6.2	1478–1528	Chamalaun and Dempsey (1978)
Reconstruction at ~1.27 Ga (after Pesonen et al., 2003)								
B	Mean Baltica	52	−24	4.0	158.0	4.0	1265	Pesonen et al. (2003)
L	MacKenzie dykes	268	12	4.0	190.0	5.0	1267+7/−3 (U-Pb)	Buchan and Halls (1990), LeCheminant and Heaman (1989)

B is Baltica, L is Laurentia, Sb is Siberia, Au is Australia (NA (SA) is North (South) Australia); Cra. Is craton; D_R, I_R refer to declination, inclination of a central reference location for each craton/continent; reference locations: B, 64°N; 28°E; L, 60°N; 275°E; Sb, 60°N, 105°E; NA, 20°S, 135°E; SA, 30°S, 135°E; Plat, Plong is latitude and longitude of the paleomagnetic pole, A_{95} is the confidence circle of the pole.

for this 1.46 Ga plate configuration for Baltica are Lat: 0°, Long: 102°, angle 79° and for Laurentia: Lat: 0°, Long: 312°, angle: −93.5°. The configuration is similar to those supported by geological data (e.g. Gorbatschev and Bogdanova, 1993).

The 1.27 Ga assembly of Baltica and Laurentia by Pesonen et al. (2003) seems to be similar to the 1.46 Ga assembly presented here. The 1.27 Ga configuration is supported also by geological and geophysical evidences, including the proposed common magma source for coeval dolerites in north and central Sweden and in Finland and basalts and dolerites in NE Greenland (e.g. Elming and Mattsson, 2001; Karlström et al., 2001). Data for Baltica comes from dolerites (Vaasa 1268 ± 13 Ma, Satakunta 1.264 ± 12 Ma and Märket 1267 ± 6 Ma; Pesonen et al., 2003) and data for Laurentia comes from MacKenzie dykes (1267 +7/−3 Ma; Buchan and Halls, 1990; LeCheminant and Heaman, 1989). In order to test paleomagnetically if Baltica and Laurentia were parts of a single continent between 1.46 and 1.27 Ga we need to test if the used ca. 1.27 Ga poles fulfil the same configuration, in this case 1.46 Ga configuration. Confidence circles of these ca. 1.27 Ga poles for Baltica and Laurentia overlap with each other and just overlap with the 1.46 Ga continental configuration poles, which means that there is paleomagnetic evidence that

1.27 Ga configuration is similar to 1.46 Ga configuration (Fig. 11).

For testing the older assembly at 1.76 Ga we use data for Laurentia from Cleaver Lake dykes (1740 +5/−4 Ma, Irving et al., 2004) and data for Baltica from key pole of Ropruchey-Shosksha sill (1770 Ma, Pisarevksy and Sokolov, 2001). Confidence circles of these ca. 1.76 Ga poles for Baltica and Laurentia overlap with each other and with the poles when using this 1.46 Ga continental configuration (Fig. 11). This gives paleomagnetic support for existence of a single continental Baltica–Laurentia block since 1.76 Ga until 1.27 Ga.

At 1.83 Ga the position of Baltica is based on a well defined pole from Haukivesi lamprophyres (1840 Ma; Neuvonen et al., 1981), and the position of Laurentia is based on the pole from the Sparrow dykes (1827 ± 4 Ma; McGlynn et al., 1974; Bostock and van Breemen, 1992), which was chosen to represent the 1.83 Ga consolidated North American continent (Pesonen et al., 2003). However, the choice of the corresponding pole for Laurentia is difficult since Laurentia may not have been consolidated yet. Using the closest approach technique Pesonen et al. (2003) placed Baltica's present NE part against Laurentia's present NNE part. This configuration of Baltica and Laurentia is also supported by coeval 2.1–1.8 Ga collisional orogens, which includes the 1.95–1.85 Ga Trans-Hudson orogen in North America, 1.9–1.8 Ga Nagssugtoqidian orogen in Greenland, and 1.9–1.8 Ga Kola-Karelia and Svecofennian orogens in Baltica (e.g. Gorbatschev and Bogdanova, 1993; Karlström et al., 2001; Zhao et al., 2004). However, the confidence circles of 1.83 Ga poles do not overlap with each other (Fig. 11), when using 1.46 Ga continental configuration. This means that there is no paleomagnetic support for the single Baltica–Laurentia block at 1.83 Ga, which may be due to the fact that the used paleomagnetic poles are not coeval in age.

8.3. Supercontinent Hudsonland

Several researchers have proposed that the Hudsonland (Columbia) configuration could have included most of the continents, which were amalgamated together along global 2.1–1.8 Ga collisional orogens and the 'maximum packing' of continental blocks was achieved at ca. 1.5 Ga ago (e.g. Meert, 2002; Pesonen et al., 2003; Rogers and Santosh, 2002; Zhao et al., 2004 and references therein). According to these authors the fragmentation of Hudsonland began ca. 1.6–1.5 Ga ago, in association with development of continental riftings and widespread anorogenic activities, which included, e.g. emplacement of anorthosite massifs, charnockite intrusions, rapakivi granites and dyke swarms, such as the Valaam monzodioritic sill. There is an ongoing debate about the time of the final break up of this assembly. It has been proposed that it is related to the ca. 1.6–1.4 Ga bimodal rapakivi magmatism (Rämö and Haapala, 2005) or to the rifting and emplacement of the 1.3–1.2 Ga mafic intrusions (e.g. SW Finland and central Sweden and the coeval mafic dike swarms in North America). Paleomagnetic results of the Valaam sill, together with coeval data from other continents, provide us a tool to study the existence and configuration of Hudsonland at ∼1.46 Ga and the possible onset of its fragmentation, since the sill has been interpreted to represent basic volcanism that followed the emplacement of the 1.65–1.50 Ga rapakivi intrusions (Rämö and Haapala, 2005; Rämö et al., 2003).

At ∼1.46 Ga, there are paleomagnetic data only for Baltica (mean of 1.46 Ga) and Laurentia (interpolated mean of 1.46 Ga; Table 6). For other continents, such as Australia (1.55–1.50 Ga) and Siberia (1.50 Ga) we have to use data with somewhat older ages (Table 6). In Fig. 12 a–d we present some of the proposed 1.5 Ga configurations for Hudsonland (Columbia). As these reconstructions are based on paleomagnetic data; we show only those continents, from which there are paleomagnetic poles for reconstructions.

The Hudsonland configuration of Pesonen et al. (2003) is presented in Fig. 12 a. This arrangement of continents is virtually the same as for Columbia by Meert (2002) (Fig. 12b), if the polarity of the poles of the Columbia are inverted. However, in both reconstructions Baltica is located at the eastern margin of Laurentia. Pesonen et al. (2003) point out that the mean paleomagnetic data of Baltica for their 1.5 Ga reconstruction are of low quality and ca. 40 Ma older than pole for Laurentia. Although, there are no paleomagnetic data for other continents of exactly the same age (1.46 Ga), the occurrence of ca. 1.6–1.5 Ga rapakivi granites in Australia (Karlström et al., 2001) support the presented position of Australia close to Baltica and Laurentia ca.1.5 Ga. Laurentia and Australia are presented as in AUSWUS model after Karlström et al. (2001), which is also supported by the younger Rodinia configuration. Siberia is in the Rodinia reconstruction (Hoffman, 1991) based on paleomagnetic data of Kuonamka dykes by Ernst et al. (2000). The position of Siberia matches also with younger configuration of Rodinia; however, the error of this particular paleomagnetic pole is of 28°.

Zhao et al. (2004 and references therein) propose a different configuration of Columbia for two reasons. Firstly, their studies are based not only on paleomagnetism but also on lithostratigraphic, tectonothermal and geochronological data, allowing them to include all con-

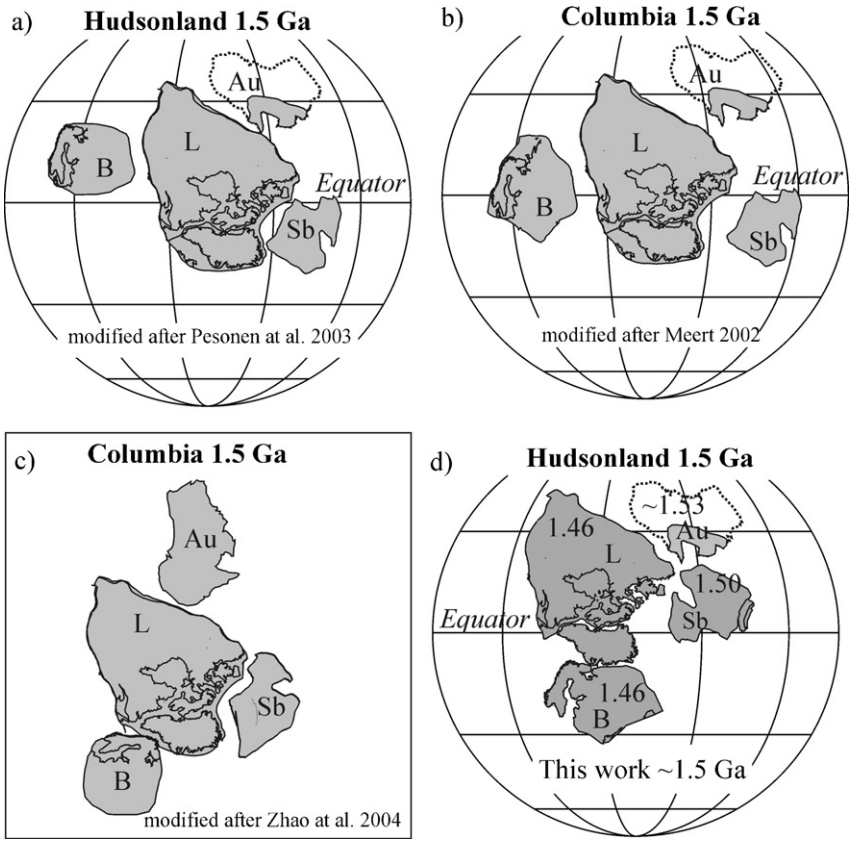


Fig. 12. The proposed Hudsonland (Columbia) supercontinent configurations showing Laurentia, Baltica, North Australia and Siberia. (a) The ca. 1.5 Ga paleomagnetic reconstruction of Hudsonland after Pesonen et al., 2003. (b) Paleomagnetic reconstruction of Columbia at ca. 1.5 Ga by Meert (2002) is similar than previous one, but used polarity is opposite. (c) Reconstruction of Columbia by Zhao et al. (2004) based on lithostratigraphic, tectonothermal, geochronological and paleomagnetic data (only continents, which have 1.5 Ga paleomagnetic data are shown). The reconstruction is rotated as a whole so that Laurentia's position is coeval with Pesonen et al. (2003) and Meert (2002). (d) Paleomagnetic reconstruction of Hudsonland based on data from this work.

tinents. Secondly, they do not provide paleolatitudinal information. For comparison, we show here only those continents for which there are reliable paleomagnetic data (Fig. 12 c). The configuration of Zhao et al. (2004) is supported by continuation of anorogenic magmatism from Laurentia to Baltica (Karlström et al., 2001). Unfortunately Zhao et al. (2004) do not provide any information about calculate to locations and rotations of the continents.

The new well defined and precisely dated paleomagnetic pole from Baltica (this work together with data from: Salmi volcanic rocks, Lake Ladoga dolerites and Tuna dykes; Table 5) allows us to provide a better reconstruction of the Baltica–Laurentia relation in Hudsonland configuration at ca. 1.46 Ga (see Section 6.2). In Fig. 12d the position of Baltica is based on the mean pole of all these units. Similarly, the position for Laurentia is based on mean pole from St. Francois Mnt, Harp Lake complex, Michicamau Intrusion and Spokane Formation from Belt Supergroup (interpolated age of 1.46 Ga; Table 6). The use of the mean poles for Baltica and Laurentia supports the idea that these continents were joined during the Mesoproterozoic. Australia's position is reconstructed as in Fig. 11a and b, while Siberia is reconstructed using the opposite polarity pole than in Fig. 11a and b.

9. Conclusions

1. Paleomagnetic studies of the 1.46 Ga mozodioritic Valaam Sill revealed the presence of two remanence components: a high coercivity well defined component **A** ($D = 43.4^\circ$, $I = -14.3^\circ$, $\alpha_{95} = 3.3^\circ$) and a viscous component related to the present earth field.
2. Although baked contact or fold tests were not possible, the overall analysis of the paleomagnetic data of the sill, of the other units around Lake Ladoga (similar ages than Valaam sill) and those of Jänisjärvi Sve-

cofennian rocks (older than Valaam sill, ca. 1.8 Ga) suggest a primary origin for component **A**. Rock magnetic and scanning electron microscope studies showed that the main carrier of the primary magnetization is titanomagnetite (PSD to MD), with varying Ti-content. Component **A** yields a new well dated (U-Pb ages of 1459 ± 3 Ma; 1457 ± 2 Ma) paleopole (Plat = 13.8°N, Plong = 166.4°E, A_{95} = 2.4°) for Baltica. The pole takes Baltica on 7°S latitude. The Mesoproterozoic ~1.5–1.3 Ga red-bed sedimentation (for example the Satakunta sandstone) supports also this low latitude location of Baltica (Rämö et al., 2005).

3. For reconstructing the 1.46 Ga positions for Baltica we used the mean pole (A) of this work together with data from Salmi volcanic rocks, Lake Ladoga dolerites and Tuna dykes. For Laurentia we used the mean pole from St. Francois Mnt, Harp Lake complex, Michicamau Intrusion and Spokane Formation from Belt Supergroup. Since the used poles for Baltica and Laurentia are not of precisely the same age and coeval it is permissible that their ovals of confidence overlap with each other and with the geographic pole. The Baltica–Laurentia fit in this case is similar to those at 1.76 and 1.27 Ga. This means that the existence of a single continental Baltica–Laurentia block since 1.76 Ga until 1.27 Ga is paleomagnetically permissible. This configuration is similar to those supported by geological data (e.g. Gorbatschev and Bogdanova, 1993).
4. We propose a new Hudsonland configuration for ca. 1.46 Ga. According to combined result from Lake Ladoga area and Tuna dikes we place N Baltica against NE Greenland, which differs for example from Meert (2002) and Pesonen et al. (2003). They placed E Baltica against E Laurentia. The new reconstruction is similar to those supported by geological data (e.g. Gorbatschev and Bogdanova, 1993). Further support for this reconstruction is continuous rapakivi magmatism (1.6–1.5 Ga) on Baltica and Laurentia (Rämö and Haapala, 2005). Australia is reconstructed followed Meert (2002) and Pesonen et al. (2003). Siberia is reconstructed using the opposite polarity option that Meert (2002) and Pesonen et al. (2003). We would need results from well-dated rock formations with better matched ages from Australia and Siberia in order to fully check position of those continents in Hudsonland reconstruction.
5. The monzodiorites of the Mesoproterozoic Valaam sill have distinctly higher densities than syenites. The syenites also show lower susceptibility and NRM values than monzodiorites as expected on the basis of different mineralogical composition, but similar Q-values.

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References

Åhäll, K.I., Connelly, J., 1998. Intermittent 1.53–1.13 Ga magmatism in western Baltica; age constraints and correlations within a postulated supercontinent. Precambrian Res. 92, 1–20.

Amantov, A., Laitakari, I., Poroshin, Ye., 1996. Jotnian and postjotnian sandstones and diabases in the surroundings of the Gulf of Finland. In: Koistinen, T. (Ed.), Explanation to the map of Precambrian basement of the Gulf of Finland and surrounding area 1:1 million. Geol. Surv. Finl., Spec. Pap. 21, 99–113.

Bostock, H.H., van Breemen, O., 1992. The timing of emplacement and distribution of the Sparrow diabase dyke swarm. District of Mackenzie, Northwest Territories. Radiogenic age and isotopic studies. Geol. Surv. Can. 92, 49–55.

Buchan, K.L., Halls, H.C., 1990. Paleomagnetism of Proterozoic mafic dyke swarms of the Canadian Shield. In: Parker, A.J., Rickwood, P.C., Tucker, D.H. (Eds.), Mafic Dykes and Emplacement Mechanism. A.A. Balkema, Rotterdam, pp. 209–230.

Buchan, K.L., Mertanen, S., Park, R.G., Pesonen, L.J., Elming, S.-Å., Abrahamsen, N., Bylund, G., 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key paleomagnetic poles. Tectonophysics 319, 167–198.

Bylund, G., 1985. Palaeomagnetism of middle Proterozoic basic intrusives in central Sweden and the Fennoscandian apparent polar wander path. Precambrian Res. 28, 283–310.

Chamalaun, F.H., Dempsey, C.E., 1978. Palaeomagnetism of the Gawler Range Volcanics and implications for the genesis of the Middleback hematite orebodies. J. Geol. Soc. Aust. 25, 255–265.

Condie, K.C., 2000. Episodic continental growth models: afterthoughts and extensions. Tectonophysics 322, 153–162.

Condie, K.C., 2004. Supercontinents and superplume events: distinguishing signals in the geologic record. Phys. Earth Planet. Sci. 146, 319–332.

Dalziel, I.W.D., 1994. Neoproterozoic-Paleozoic geography and tectonics: review, hypotesis and environmental speculation. Geol. Soc. Am. Bull. 109, 16–42.

Day, R., Fuller, M.D., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain size and composition dependence. Phys. Earth Planet. Int. 13, 260–267.

Dragoni, M., Lanza, R., Tallarico, A., 1997. Magnetic anisotropy produced by magma flow: theoretical model and experimental data from Ferrar dolerite sills (Antarctica). Geophys. J. Int. 128, 230–240.

Dunlop, D.J., 1974. Thermal enhancement of magnetic susceptibility. J. Geophys. 40, 439–451.

Dunlop, D.J., 1983. Determination of domain structure in igneous rocks by alternating field and other methods. Earth Planet. Sci. Lett. 63, 353–367.

Dunlop, D.J., 2002. Theory and application of the Day-plot (Mrs/Ms versus Hcr/Hc) 1. Theoretical curves and tests using titanomagnetite data. J. Geophys. Res. 107, NO. B3, 10.1029/2001JB000486.

Dunlop, D.J., Özdemir, Ö., 1997. Rock Magnetism, Fundamentals and Frontiers. Cambridge University Press, Cambridge, 573 p.

Elming, S.-Å., Mattsson, H., 2001. Post Jotnian basic intrusions in the Fennoscandian shield, and the break up of Baltica from Laurentia: a paleomagnetic and AMS study. Precambrian Res. 108, 215–236.

Elming, S.-Å., Mikhailova, N.P., Kravchenko, S., 2001. Paleomagnetism of Proterozoic rocks from Ukrainian shield: new tectonic reconstructions of the Ukrainian and Fennoscandian shields. Tectonophysics 339, 19–38.

Elston, D.P., Bressler, S.L., 1980. Paleomagnetic poles and polarity zonation from the Middle Proterozoic Belt Supergroup, Montana and Idaho. J. Geophys. Res. 85, 339–355.

Emslie, R.F., Irving, E., Park, J.K., 1976. Further paleomagnetic results from the Michikamau intrusion, Labrador. Can. J. Earth Sci. 13, 1052–1057.

Ernst, R.E., Buchan, K.L., Hamilton, M.A., Okrugin, A.V., Tomshin, M.D., 2000. Integrated paleomagnetism and U-Pb geochronology of mafic dikes of the eastern Anabar Shield region, Siberia: implications for Mesoproterozoic paleolatitude of Siberia and comparison with Laurentia. J. Geol. 108, 381–401.

Fisher, R., 1953. Dispersion of sphere. Proc. R. Soc. Lond. A 217, 293–305.

Gorbatschev, R., Bogdanova, S., 1993. Frontiers in the Baltic Shield. Precambrian Res. 64, 3–21.

Harlan, S.S., Geissman, J.W., 1998. Paleomagnetism of the Middle Proterozoic Electra Lake Gabbro, Needle Mountains, southwestern Colorado. J. Geophys. Res. 103, 15497–15507.

Harlan, S.S., Snee, L.W., Geissman, J.W., 1994. Paleomagnetism of the Middle Proterozoic Laramie anorthosite complex and Sherman granite, southern Laramie Range, Wyoming and Colorado. J. Geophys. Res. 99, 17997–18020.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwana inside out? Science 252, 1409–1412.

Hrouda, F., 1982. Magnetic anisotropy of rocks and its application in geology and geophysics. Geophys. Surv. 5, 37–82.

Irving, E., Baker, J., Hamilton, M., Wynne, P.J., 2004. Early Proterozoic geomagnetic field in western Laurentia: implications for paleolatitudes, local rotations and stratigraphy. Precamrbian Res. 129, 251–270.

Irving, E., Emslie, R.F., Park, J.K., 1977. Paleomagnetism of the Harp Lake Complex and associated rocks. Can. J. Earth Sci. 14, 1187–1201.

Karlström, K.E., Åhäll, K.-I., Harlan, S., Williams, M.L., McLelland, J., Geissman, J.W., 2001. Long-lived (1.8–1.0 Ga) convergent orogen in southern Laurentia, its extensions to Australia and Baltica, and implications for refining Rodinia. Precambrian Res. 111, 5–30.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Koistinen, T., Stephens, M.B., Bogatchev, V., Nordgulen, Ø., Wennerström, M., Korhonen, J., 2001. Geological map of the Fennoscandian Shield, scale 1:2 000 000. Geol. Surv. (Finland, Norway and Sweden and the North-West Department of Natural Resources of Russia).

Laitakari, I., Koistinen, T., Virransalo, P., 1996. Principles and sources for the basement map, Gulf of Finaldnand surrounding area. In: Koistinen, T. (Ed.), Explanation to the map of Precambrian basement of the Gulf of Finland and surrounding are 1:1 mill. Geol. Surv. Finl., Spec. Pap. 21, 9–13.

LeCheminant, A.N., Heaman, L.M., 1989. MacKenzie igneous events, Canada: middle Proterozoic hotspot magmatism associated with ocean opening. Earth Planet. Sci. Lett. 96, 38–48.

Leino, M.A.H., 1991. Paleomagneettisten tulosten monikomponenttianalyysi pienimmän neliösumman menetelmällä (multicomponent analysis of paleomagnetic data by least-squares method). Geolo. Surv. Finl. Report Q29.1/91/2. Laboratory for Paleomagnetism, Department of Geophysics, 15 pp. (in Finnish).

Lubnina, N., Mertanen, S., Vasilieva, T., 2005. Paleomagnetism of middle Riphean dykes from the Ladoga Lake region of Northern Karelia. In: Wingate, M.T.D., Pisarevsky, S.A. (Eds.), Supercontinents and earth evolution Symposium 2005, Fremantle, Western Australia at the Maritime Museum, Victoria Quay. Geol. Soc. Aust., 75 (Abstracts 81).

Lubnina, N., Cecys, A., Söderlund, U., 2007. Paleomagnetic studies on the Mesoproterozoic dykes in Central Sweden: preliminary results (1607-7962/gra/EGU2007-A-08308).

McGlynn, J.C., Hanson, G.N., Irving, E., Park, J.K., 1974. Paleomagnetism and age of Nonacho Group sandstones and associated Sparrow dikes, district of Mackenzie. Can. J. Earth Sci. 11, 30–42.

Meert, J.G., 2002. Paleomagnetic evidence for a Paleo-Mesoproterozoic supercontinent Columbia. Gondwana Res. 5, 207–215.

Meert, J.G., Stuckey, W., 2002. Revisiting the paleomagnetism of the 1.476 Ga St. Francois Mountains igneous province. Missouri. Tectonics 21 (10.1029/2000TC001265).

Merrill, R.T., McElhinny, M.W., 1983. The Earth's Magnetic Field: Its History, Origin, and Planetary Perspective. Academic Press, London, p. 401.

Mertanen, S., 2005. Paleo- and Mesoproterozoic dyke swarms at Lake Ladoga area, NW Russia-paleomagnetic studies. In: Vuollo, J., Mertanen, S. (Eds.), Fifth International dyke Conference. Pohtimolampi, Finland, 31 July–3 August, p. 33 (Abstracts and Programme).

Mertanen, S., Pesonen, L.J., 1994. Preliminary results of a paleomagnetic and rock magnetic study of the Proterozoic Tsuomasvarri intrusion, northern Fennoscandia. Precambrian Res. 69, 25–50.

Mertanen, S., Airo, M.-L., Elminen, T., Niemelä, R., Pajunen, M., Wasenius, P., Wennerstöm, M., 2006. Paleomagnetic evidence for Mesoproterozoic-Paleozoic reactivation of the Paeozoic crust in southern Finland. Bull. Geol. Soc. Finl. (Special issue 1), 102.

Neuvonen, K.J., Korsman, K., Kouvo, O., Paavola, J., 1981. Paleomagnetism and age relations of the rocks in the main sulphide ore belt in central Finland. Bull. Geol. Soc. Finl. 53, 109–133.

Pesonen, L.J., 1998. Paleomagnetism of a sample from the Valaam sill, Lake Ladoga, Russia. Geol. Surv. Finl. Report Q29.1/98/1. Laboratory for Paleomagnetism, 6 pp.

Pesonen, L.J., Elming, S.-Å., Mertanen, S., Pisarevsky, S., D'Agrella-Filho, M.S., Meert, J.G., Schmidt, P.W., Abrahamsen, N., Bylund, G., 2003. Paleomagnetic configuration of continents during the Proterozoic. Tectonophysics 375, 289–324.

Pisarevksy, S.A., Sokolov, S.J., 2001. The magnetostratigraphy and a 1780 Ma paleomagnetic pole from the red sandstones of Vazhinka River section, Karelia, Russia. Geophys. J. Int. 146, 531–538.

Rämö, O.T., Mänttäri, I., Vaasjoki, M., Upton, B.G.J., Sviridenko, L., 2001. Age and significance of Mesoproterozoic CFB magmatism, Lake Ladoga region, NW Russia. Boston 2001: A Geo-Odyssey. GSA Annual Meeting and Exposition Abstracts, November 1–10. Geol. Soc. Am. 33 (6), p. A-L 139 (Abstracts with Programs).

Rämö, O.T., McLemore, V.T., Hamilton, M.A., Kosunen, P.J., Heizler, M., Haapala, I., 2003. Intermittent 1630–1220 Ma magmatism in central Mazatzal province: new geochronologic piercing point and some tectonic implications. Geology 31, 335–338.

Rämö, O.T., Haapala, I., 2005. Rapakivi granites. In: Lehtinen, M., Nurmi, P.A., Rämö, O.T. (Eds.), Precambrian geology of Finland—Key to the Evolution of the Fennoscandian Shield. Elsevier B.V., Amsterdam, pp. 533–562.

Rämö, O.T., Mänttäri, I., Kohonen, J., Upton, B.G.J., Luttinen, A.V., Lindqvist, V., Lobaev, V.M., Cuney, M., Sviridenko, L.P., 2005. Mesoproterozoic CFB magmatism in the Lake Ladoga basin, Russian Karelia. In: Vuollo, J., Mertanen, S. (Eds.), Fifth International dyke Conference. Pohtimolampi, Finland, 31 July–3 August, pp. 41–42 (Abstracts and Programme).

Rogers, J.J.W., Santosh, M., 2002. Configuration of Columbia, a Mesoproterozoic supercontinent. Gondwana Res. 5, 5–22.

Salminen, J., Donadini, F., Pesonen, L.J., Masaitis, V.L., Naumov, M., 2006. Paleomagnetism and petrophysics of the Jänisjärvi impact structure, Russian Karelia. Meteorit. Planet. Sci. 41, 1853–1870.

Shcherbakova, V.V., Pavlov, V.E., Shcherbakov, V.P., Neronov, I., Zemtsov, V.A., 2006. Paleomagnetic studies and estimation of geomagnetic paleointensity at the Early/Middle Riphean boundary in rocks of the Salmi formation (north Ladoga area). Izvestiya, Phys. Solid Earth 42, 233–243.

Söderlund, U., Isachsen, C.E., Bylund, G., Heaman, L.M., Patchett, P.J., Vervoort, J.D., Andersson, U.B., 2005. U-Pb baddelyite ages and Hf, Nd isotope chemistry constraining repeated mafic magmatism in the Fennoscandian Shield from 1.6 to 0.9 Ga. Contrib. Mineral. Petrol. 150, 174–194.

Tanaka, H., Idnurm, M., 1994. Palaeomagnetism of Proterozoic mafic intrusions and host rocks of the Mount Isainlier, Australia: revisited. Precambrian Res. 69, 241–258.

Tauxe, L., 2002. The PMAG software package Online Documentation for use with Paleomagnetic Principles and Practice. Kluwer Academic Publishers, p. 113.

Torsvik, T., Smethurst, M., 1999. Plate tectonic modelling: virtual reality with GMAP. Comput. Geosci. 25, 395–402.

Vitorello, I., Van der Voo, R., 1977. Late Hadrynian and Helikian pole positions from the Spokane formation, Montana. Can. J. Earth Sci., 67–73.

Vaasjoki, M., Korsman, K., Koistinen, T., 2005. Overwiev. In: Lehtinen, M., Nurmi, P.A., Rämö, O.T. (Eds.), Precambrian geology of Finland—Key to the Evolution of the Fennoscandian Shield. Elsevier B.V., Amsterdam, pp. 1–18.

Zhao, G., Sun, M., Wilde, S., Li, S., 2003. Assembly, accretion and breakup of the Paleo-Mesoproterozoic Columbia supercontinent: records in the North China craton. Gondwana Res. 6, 417–434.

Zhao, G., Sun, M., Wilde, S.A., Li, S., 2004. A Paleo-mesoproterozoic supercontinent: assembly, growth and breakup. Earth Sci. Rev. 67, 91–123.

Zijderveld, J.D.A., 1964. AC demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, New York, pp. 254–286.